

# What is a Vector Space?

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# What are some examples?

$$V = \mathbb{R}^n, F = \mathbb{R}.$$

## What are some examples?

$V = \mathbb{R}^n, F = \mathbb{R}$ . See Slide 8 of Lecture slide 3 for properties 2-5 and 7-10.

For properties 1 and 2, we have the definitions of vector addition and scalar multiplication that guarantees this!

## More Examples

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$$\begin{aligned} 1. \quad & (a_2x^2 + a_1x + a_0) + (b_2x^2 + b_1x + b_0) = \\ & (a_2 + b_2)x^2 + (a_1 + b_1)x + (a_0 + b_0) \end{aligned}$$

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- $f(x) + g(x) = g(x) + f(x)$
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$$3. (f(x) + g(x)) + h(x) = f(x) + (g(x) + h(x))$$

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## More Examples

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10.  $1f(x) = f(x)$

## Is this a Vector Space?

$V = \mathbb{R}^3, F = \mathbb{R}$  using standard scalar multiplication but  $\mathbf{u} + \mathbf{v} = \begin{bmatrix} u_1 + v_2 \\ u_2 + v_2 \\ u_3 + v_3 \end{bmatrix}$

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Thus, we see that  $\mathbf{w} = \mathbf{0}$



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## Vector Space Practice

Work with your neighbors to determine if the following spaces are vector spaces

$V = \mathbb{R}^3$ ,  $F = \mathbb{R}$  with the usual vector addition

and  $c\mathbf{u} = \begin{bmatrix} -cu_1 \\ -cu_2 \\ -cu_3 \end{bmatrix}$ .

$V = \mathbb{R}^{3 \times 3}$ ,  $F = \mathbb{R}$  with the standard operations.

# Vector Subspaces

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A subspace of a vector space  $V$  is a subset  $H$  of  $V$  ( $H \subseteq V$ ) that has the following properties (using the same vector addition, scalar multiplication, and  $F$ )

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3.  $H$  is closed under scalar multiplication: for each  $c \in F$  and  $\mathbf{v} \in H$ , we have  $c\mathbf{v} \in H$ .

## Is it a Subspace?

Determine with your neighbors if each of the following sets are subspaces of  $V = \mathbb{R}^3$ .

$$H = \left\{ \mathbf{v} \in \mathbb{R}^3 \mid \mathbf{v} = \begin{bmatrix} 3a + b \\ a + 5 \\ 2a - 5b \end{bmatrix} \text{ for } a, b \in \mathbb{R} \right\} \qquad H = \left\{ \mathbf{v} \in \mathbb{R}^3 \mid \mathbf{v} = \begin{bmatrix} a \\ 0 \\ b \end{bmatrix} \text{ for } a, b \in \mathbb{R} \right\}$$

# Spanning Sets and Subspaces

Let  $\mathbf{v}_1, \dots, \mathbf{v}_p$  denote a set of  $p$  vectors in  $V$ . Then  $\text{Span}(\mathbf{v}_1, \dots, \mathbf{v}_p)$  is a subspace of  $V$ .

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2. Given any subspace  $H$  of  $V$ , a spanning set for  $H$  is a set  $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$  of vectors in  $H$  such that  $H = \text{Span}(\mathbf{v}_1, \dots, \mathbf{v}_p)$

## Example

Determine if  $H = \left\{ A \in \mathbb{R}^{3 \times 3} \mid A = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}, a + b + c = 0 \right\}$  is a subspace of  $\mathbb{R}^{3 \times 3}$  and if so, give a spanning set for  $H$ .

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So,  $H$  is a subspace of  $\mathbb{R}^3$ !

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## Example continued pt. 3

$$a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3 = a \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} + b \begin{bmatrix} -1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{bmatrix} + c \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

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# Spanning and Independent List of Correct Size is a Basis

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